Prompting Field Guide

Techniques for better and more truthful outputs, in the vocabulary of the study guide. SMALL CAPITALS mark a study-guide entry you can look up.

What a prompt can and cannot do. A prompt is a TECHNIQUE. It changes the text in front of the model and nothing in the WEIGHTS. Three facts explain why every technique below works:

- The model writes **one token at a time**, each drawn from a SOFTMAX over scores. A prompt works by changing which continuation is likely.
- Through ATTENTION, every token it writes can look at everything before it and nothing after it (the CAUSAL MASK). What you put in the prompt, and in what order, is what it has to work with.
- **Fluent is not true.** A HALLUCINATION reads exactly like a correct answer. "It sounds right" is the one test it always passes.

1 · Say what you want constraints

A chat box affords typing anything and signals almost nothing. The prompt is where you supply the CONSTRAINTS the interface left out.

- **Name the task, the audience, the format and the length.**
- **Say what a good answer contains**, not that it should be good.
- **Say what to do when the input does not support an answer.**
- **Separate instructions from material.** Label the parts. Put a long document first and the question after it.
- **Ask for a format you can check:** one label from a list, a number, named fields.

```
Weak:   Summarize this article well.
Better: Summarize the article below in three
        sentences for a first-year student. Name the
        disagreement, not the topic. If the article
        does not say who funded the study, write
        "funding not stated".
```

2 · Set the frame system prompt, role, context

- **System prompt.** Standing instructions that apply to every turn of a conversation: who the answers are for, what the model must always and never do, when to decline, and the output format. Write it once, keep it in a file, and change it on purpose.
- **Role.** "You are a reference librarian" reliably changes voice and vocabulary. **Limit:** *whether it changes accuracy depends on the task. Test it; do not assume it.*
- **Context.** Give the model the facts it cannot have: your document, your definitions, today's date, anything after its training data ends.

```
You answer questions about the course syllabus
pasted below, for students in the course.
Use only the syllabus. Quote the sentence your
answer comes from. If the syllabus does not
answer the question, reply "not in the syllabus"
and suggest emailing the instructor.
```

3 · Show it zero-shot and few-shot

- ZERO-SHOT: the instruction alone. Start here.
- FEW-SHOT: two or three worked examples before the real input. Examples teach format and habit faster than description does.
- **Choose examples that differ from each other**, and include one where the right answer is "no" or "not stated".

Limit: *a mistake in an example is taught too, and three similar examples teach the model that every input is like them.*

4 · Give it room chain of thought and relatives

- CHAIN OF THOUGHT: ask for the steps before the answer. Zero-shot: add "Let's think step by step." Few-shot: show a worked example with its steps (Wei et al.).
- **Break the job up.** One prompt per step, with the output of one as the input to the next, beats one prompt that does everything.
- **Ask more than once.** Run the same prompt several times in new chats and compare. Agreement is weak evidence; disagreement is a warning.

Limit: *the steps are text the model produced, not a record of how it got the answer (FAITHFULNESS). In the Week 6 lab the small models wrote fluent, illogical chains. It helps most on problems like the ones the model has seen and least OUT OF DISTRIBUTION.*

Read more: Boonstra, *Prompt Engineering* (Week 4) · Schulhoff et al., *The Prompt Report* (Week 5) · Wei et al. on chain of thought (Week 6) · Lin, Hilton & Evans, *TruthfulQA* (Week 7) · `PROMPTING.md` in your project repository.

5 · Alignment: keeping the model on the truth Week 7

TruthfulQA showed models repeating popular myths because the myth is the likely continuation: an IMITATIVE FALSEHOOD. These are the moves that work against that, and against invention generally. None of them makes a model know something it does not.

The risk	What to put in the prompt	Why it helps, and what to watch
It repeats the popular answer	Ask for the careful voice: *"Interpret the question literally. Answer as a careful reference source would. Do not repeat common myths."*	Makes a careful source the likely voice. The paper's best score came from its "helpful" prompt, which did this.
It has no way to say "I don't know"	Give it one: *"If you are not certain, reply 'I have no comment.'"* or *"If it is not in the text, write 'not stated'."*	A model with no permitted way out invents one. **Count how often it takes it:** in the lab this prompt doubled the score, mostly by refusing eight of twenty questions.
It invents facts, quotations or sources	Supply the source text and require a quotation for each claim. If you ask for references, look every one up.	A reference is a pattern (authors, year, journal) and easy to produce whether or not the paper exists.
Your question contains a false premise	Ask neutrally, or add: *"First say whether the question's assumption is true."*	A model tends to answer the question it was asked. "Why did X win the prize?" gets a reason even if X never won.
The fact is obscure, local or recent	Paste the facts in. Do not rely on the model's memory.	Little was written about it, so there is little to be right from. This is what retrieval (Week 10) automates.
Earlier turns are steering it	Start a new chat for every test.	The model can see the whole conversation, including your reaction to its last answer.
You want it to check itself	In a new chat, give it the answer and the source and ask what in the answer the source does not support.	Checking against a text is a task it can do. Asking "are you sure?" mostly tests whether it will change its mind.

6 · Measure it Thursday's lab

You cannot claim a prompt is better without a way to measure it.

1. Keep an EVALUATION SET: real inputs, each with GROUND TRUTH and a source. Include inputs you expect to fail and ones where the right answer is "no".
2. **Change one thing at a time**, and record the count before and after.
3. Choose the MARKING RULE first, because it matches your own marks. Then look at the score.
4. Report TRUTHFUL AND INFORMATIVE together. A model that says nothing false by saying nothing has not improved.
5. With twenty items, one item is five points. Name the items that changed, in both directions.
6. A second person's marks will differ from yours (INTER-ANNOTATOR AGREEMENT). Where they differ, fix the item or the rule.
7. When you find a new failure, add it to the set **before** you fix it.

7 · Symptom → what to try

Vague or generic	Add the audience, the length, and what a good answer contains.
Wrong format	Show one example of the format. Ask for named fields.
Ignores an instruction	Say it once, as what to do ("write in paragraphs") instead of what not to do. Separate it from the pasted material.
Makes things up	Section 5: a way to say no, the source text, a required quotation.
Wrong on a multi-step problem	Chain of thought, or split it into steps you can check.
Different every time	Chat tools sample with some TEMPERATURE. Run it several times and report the spread.
Works on your example only	That example was one item. Add the failures to your set.

Before you trust an output

- Could I check this against a source, and did I?
- Did the prompt allow "I don't know"?
- Did I ask in a new chat, more than once?
- Is this a prompt that works, or a prompt I can defend? Write the decision in an ADR.